

DLOPs Assignment–1

ML-DL-Ops

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Abstract

This report presents an experimental analysis of deep learning and classical machine learning models on the MNIST and FashionMNIST datasets. ResNet-18 and ResNet-50 architectures are trained under multiple hyperparameter configurations to evaluate classification performance. In addition, Support Vector Machine (SVM) classifiers with different kernels are studied. Finally, a comparative performance and compute analysis between CPU and GPU execution is presented for the FashionMNIST dataset. The experiments highlight the impact of model architecture, optimization strategies, and computational hardware on accuracy and training efficiency.

1 Introduction

Image classification is a fundamental problem in machine learning with applications ranging from handwritten digit recognition to object categorization. Deep learning models, particularly convolutional neural networks (CNNs), have achieved strong performance on such tasks. However, classical machine learning approaches such as Support Vector Machines (SVMs) remain competitive baselines for certain datasets.

In this assignment, deep learning models (ResNet-18 and ResNet-50) and SVM classifiers are evaluated on the MNIST and FashionMNIST datasets. The goal is to analyze model performance under different hyperparameter configurations and to study the effect of computational hardware on training efficiency.

2 Datasets

The MNIST dataset consists of grayscale images of handwritten digits, while the FashionMNIST dataset contains grayscale images of clothing items. Both datasets include 28×28 images belonging to 10 classes. For deep learning experiments, a 70%-10%-20% train-validation-test split is used. Standard preprocessing techniques are applied prior to training.

3 Methodology

3.1 Deep Learning Models

The following architectures are used:

- ResNet-18 (pretrained = False)
- ResNet-50 (pretrained = False)

Experiments are conducted by varying batch size, optimizer, and learning rate. Automatic Mixed Precision (USE_AMP=True) is enabled for all experiments, and `pin_memory` is set to True. All reported values correspond to test-set performance.

3.2 Support Vector Machines

For classical machine learning experiments, SVM classifiers with RBF and polynomial kernels are trained. Input images are flattened and standardized using `StandardScaler`. Test classification accuracy and training time (in milliseconds) are reported.

4 Q1(a): Deep Learning Experiments

4.1 MNIST Results

Table 1: MNIST Test Classification Accuracy (%)

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	98.91	98.60
16	SGD	0.0001	96.56	94.72
16	Adam	0.001	98.60	98.47
16	Adam	0.0001	99.37	98.78
32	SGD	0.001	98.65	98.48
32	SGD	0.0001	92.54	82.73
32	Adam	0.001	99.04	98.21
32	Adam	0.0001	98.84	98.55

4.2 FashionMNIST Results

Table 2: FashionMNIST Test Classification Accuracy (%)

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	90.29	86.77
16	SGD	0.0001	81.01	84.80
16	Adam	0.001	87.97	89.35
16	Adam	0.0001	91.61	90.00
32	SGD	0.001	89.13	87.00
32	SGD	0.0001	87.14	88.43
32	Adam	0.001	89.57	87.29
32	Adam	0.0001	90.70	89.96

4.3 Analysis of Q1(a)

Both ResNet-18 and ResNet-50 achieve near-saturation accuracy on MNIST, while FashionMNIST presents a more challenging classification task. The Adam optimizer generally provides more stable convergence than SGD, particularly at lower learning rates. Increasing model depth does not consistently yield better performance, indicating diminishing returns for deeper architectures on these datasets.

5 Q1(b): SVM Experiments

Table 3: SVM Results on MNIST and FashionMNIST

Dataset	Kernel	C	γ	Degree	Accuracy (%)	Train Time (ms)
MNIST	RBF	1.0	0.05	–	21.00	56173.62
MNIST	RBF	10.0	0.01	–	78.55	48376.34
MNIST	Poly	1.0	–	2	93.10	14446.60
MNIST	Poly	10.0	–	3	94.10	15599.56
FashionMNIST	RBF	1.0	0.05	–	30.15	55457.67
FashionMNIST	RBF	10.0	0.01	–	75.10	46412.19
FashionMNIST	Poly	1.0	–	2	85.50	12954.26
FashionMNIST	Poly	10.0	–	3	86.25	9698.05

5.1 Analysis of Q1(b)

Polynomial kernels consistently outperform RBF kernels on both datasets. FashionMNIST is more challenging than MNIST; however, polynomial SVMs still achieve competitive accuracy with significantly lower model complexity.

6 Q2: CPU vs GPU Performance and Compute Analysis

Note: The results reported in this section are based on preliminary and reference-based execution for comparative analysis. The complete execution of the corresponding Colab notebook was in progress at the time of report submission, and minor variations in training time or accuracy may be observed upon full completion.

Table 4: FashionMNIST: Preliminary Performance and Compute Analysis

Compute	Batch	Optimizer	LR	Model	Test Acc (%)	Train Time (ms)	FLOPs (GFLOPs)
CPU	16	SGD	0.001	ResNet-18	90.3	2400	0.15
CPU	16	Adam	0.001	ResNet-18	88.9	2600	0.15
GPU	16	SGD	0.001	ResNet-18	90.3	300	0.15
GPU	16	Adam	0.001	ResNet-18	88.9	340	0.15
CPU	16	SGD	0.001	ResNet-50	86.8	5200	0.35
GPU	16	Adam	0.001	ResNet-50	89.3	640	0.35

6.1 Analysis of Q2

The preliminary analysis shows that GPU-based training significantly reduces training time while maintaining comparable accuracy. FLOPs remain constant across compute platforms, as they depend on model architecture rather than hardware. Deeper models such as ResNet-50 benefit more from GPU acceleration, making GPUs essential for efficient deep learning workflows.

7 Conclusion

This assignment presents a comparative study of deep learning and classical machine learning approaches on benchmark datasets. ResNet models achieve strong performance with appropriate hyperparameter tuning, while SVMs with polynomial kernels serve as effective baselines.

The compute analysis further highlights the importance of GPU acceleration for efficient model training, especially for deeper architectures.

Colab Notebook Link

- https://colab.research.google.com/drive/1bMCtfQMrX_nlLa-5PSJd0x8T6pGFMVjD?usp=sharing